

Prompt Optimization and Evaluation

Let's begin the discussion by answering a few questions.

Prompt Optimization and Evaluation Quiz

Which evaluation metric is robust to paraphrases because it compares meaning using contextual, embedding-based similarity?

A

BLEU

B

ROUGE

C

Exact Match

D

BERTScore

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Evaluation Metrics - Lexical

BLEU

Measures **n-gram** overlap between candidate and reference with a brevity penalty.

Introduced and widely used for machine translation tasks

Favors exact word overlap, so it underscores good paraphrases

ROUGE

Recall-focused metric checking how much of the reference's n-grams appear in the candidate.

Standard choice for evaluating summaries.

Variants (ROUGE-1, -2, -L) capture unigram, bigram, and sequence-level overlap.

Exact Match

Binary metric: score 1 if output string matches reference exactly, else 0.

Common in tasks like extractive QA or structured prediction with fixed answers.

Too strict for open-ended generation where many valid phrasings exist.

Lexical metrics reward output that LOOKS like the reference - not output that MEANS the same thing. Useful as baselines, never primary.

Evaluation Metrics - Semantic

BERTScore

Contextual, embedding-based similarity between candidate and reference using BERT

Compares meaning rather than exact words

More robust to paraphrases than lexical metrics, but can be sensitive to choice of model and language coverage

COMET

COMET (Crosslingual Optimized Metric for Evaluation of Translation) is a Neural evaluation model (usually for Machine Translation)

It scores quality using source, hypothesis, and sometimes reference.

Trained on human judgments, it better aligns with human perception of adequacy and fluency than pure overlap-based metrics

LLM-as-a-Judge

Reasoning-based evaluation using a large language model to score outputs based on a natural language rubric.

Evaluates complex dimensions like logic, instruction-following, and tone without requiring a reference

Highly aligned with human reasoning and nuance, but susceptible to biases which might lead to unfair or inaccurate scores

Evaluation Metrics - Lexical vs Semantic

Paraphrase

Reference: "Revenue grew 12% YoY."

Output: "YoY revenue increased by twelve percent."

Lexical: LOW (few n-grams)

Semantic: HIGH (same meaning)

Added word

Reference: "The deployment succeeded."

Output: "The deployment has not succeeded."

Lexical: HIGH (n-gram overlap)

Semantic: OPPOSITE (meaning is inverted)

Exact match

Reference: "Paris is capital of France."

Output: "Paris is capital of France."

Lexical: PERFECT

Semantic: PERFECT

Prompt Optimization and Evaluation Quiz

A prompt described as "stable but sensitive" indicates what two distinct diagnostics about its production readiness?

A

The model is too random, but the output is consistent across runs

B

Output is consistent across repeated runs, but small semantically equivalent tweaks cause significant change

C

The prompt is brittle, but the model + parameter combination is reliable

D

The prompt passed quality checks in a single run, but variance is unknown

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Prompt Stability vs Prompt Sensitivity

LLM outputs vary both across repeated runs and with small prompt changes
Both must be measured.



Stability

Method:

- Run SAME prompt multiple times
- Measure output consistency across runs

Analysis:

- Low stability → model unreliable at this temperature
- Diagnoses: model + parameter combination



Sensitivity

Method:

- Make small, semantically equivalent tweaks
- Measure how much output changes

Analysis:

- High sensitivity → prompt is brittle
- Diagnoses: prompt design

A prompt can be stable but sensitive.
You need BOTH to know it's production-ready.

Prompt Optimization and Evaluation Quiz

A team notices that their model's performance plateaus despite adding more training data. Upon inspection, different annotators have labelled the same examples differently. What does this most likely indicate according to IAA principles?

A

The model has reached its architectural limit and needs a larger architecture

B

Low IAA means the rubric is unclear or the task is ambiguous, and the model is learning noise rather than signal

C

The gold set is too small and needs more examples added

D

The annotators need more domain expertise before labelling

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Inter-Annotator Agreement (IAA)

If two trained humans can't agree on a label, no model trained on those labels, and no evaluator scoring against them can be trusted.

Data Quality Signal

Is the gold set actually gold?

- ▶ High IAA = labels are reliable
- ▶ Low IAA = rubric unclear or task is ambiguous
- ▶ Noisy labels causes the model to learn the noise

Rubric Diagnostic

Disagreement points at the fix

- ▶ Low-IAA items expose missing edge-case rules
- ▶ Forces a real definition of each label
- ▶ Improve the rubric, not the annotators

Model Ceiling

Humans cap what a model can achieve

- ▶ A model can't reliably exceed human agreement
- ▶ Reporting accuracy without IAA is misleading

Measure IAA BEFORE you trust the dataset, NOT after you've trained on it

IAA for Diagnostics

Low agreement on easy cases

Rubric is underspecified. Annotators are applying different implicit criteria.

Low agreement on hard cases only

Usually expected and occurs when the data is genuinely ambiguous. Document the ambiguity and don't force consensus.

Drift over time

Early agreement high, later low. Signals training wear-off or changing interpretation.

Systematic bias

One annotator consistently scores higher/lower. Calibration needed.

Disagreement usually means the rubric or example is the problem - not the annotators.

DSPy in the Real World

DSPy has moved beyond research: it now runs in production LLM pipelines at major enterprises across industries.

Retail & E-commerce

- Shopify : DSPy + GEPA optimize metadata extraction across every shop on the platform
- Sephora : structured, evaluation-driven GenAI workflows in production

Travel, Cloud & Enterprise SaaS

- JetBlue (on Databricks) : RAG chatbots and customer-feedback classification
- AWS : prompt migration from larger to smaller Amazon Nova models
- Dropbox, VMware, Moody's : ranking, evaluation, and enterprise LLM pipelines

Developer Platforms

- Replit : code-generation and agentic coding workflows
- Databricks : native DSPy + MLflow integration for deployment at scale

Case Study: JetBlue × Databricks

JetBlue replaced manually-tuned LangChain prompts with DSPy modules for its production chatbot stack — turning prompt tuning into an automated, measurable process.

From manual prompts to DSPy modules

- Multi-stage pipeline: a single wrong word in a prompt could break the whole deployment
- Each LLM call defined as a DSPy “signature”, e.g. `dspy.ChainOfThought`(“context, question -> query”)
- Custom metrics (retrieval quality, answer quality) drive DSPy’s optimizers directly, no more hand-tuning
- Deployed via MLflow PyFunc to Databricks Model Serving endpoints

Applications & measured results

- RAG-powered chatbot for predictive maintenance queries
- Revenue-driving customer-feedback classification pipeline
- Role-based document upload system for different user groups
- RAG chatbot deployment ran 2× faster than the prior LangChain deployment

Lesson: treat prompts as tunable parameters, not hand-written strings — and optimize them against real metrics.

Case Study: Shopify, Dropbox & AWS

Shopify Metadata extraction, at platform scale

- DSPy + the GEPA optimizer generate structured product metadata across every shop on Shopify
- Reported ~550× reduction in cost versus the prior approach

Dropbox Evaluation infrastructure

- Used DSPy to optimize the “Dash” relevance judge
- Powers ranking quality checks and evaluation for search/retrieval features

AWS Model & cost optimization

- DSPy used to migrate prompts from larger to smaller Amazon Nova models
- Similar task performance retained at a fraction of the inference cost

Across every case: DSPy turns prompt engineering into a systematic, evaluation-driven optimization loop — not guesswork.

Q&A



Power Ahead !

